Due: Thursday, October 12$^{\text{th}}$ at the beginning of class.

Read Section 1.2 and 1.3 of Levin.

(1) A group of 15 dancers are creating a routine. No explanation required.
   (a) How many ways can all the dancers line up on stage?
   (b) If they want to have a section of their routine with a group of 6 dancers are on stage, how many possible different groups can be made?
   (c) Another part of the routine calls for 10 dancers to be lined up. How many different ways can they pick dancers to achieve this?
   (d) Finally, there is a moment where 10 dancers will be lined up in two lines of five each, one line on the left and on the right of the stage. How many different ways can they pick dancers to achieve this?

(2) A pizza shop has four sizes of pizza, three kinds of sauce and nine different toppings. No explanation required.
   (a) How many ways are there to pick a pizza with one topping?
   (b) How many ways are there to pick a pizza with two different toppings?
   (c) How many ways are there to pick a pizza with two different toppings or doubling up one topping?
   (d) How many ways are there to pick a pizza with three different toppings?
   (e) How many different pizzas are there with any number of toppings?

(3) Let's talk about pin codes, like the ones used for banks and phone companies. A pin code is a usually 4 digit number composed of the numbers 0 - 9. For example 0011 is a pin code and it is not the same as the number 11. **Explain** each of your answers, using complete sentences.
   (a) You have a friend Alice (that is not very good at discrete math) who uses a 4 digit pin number for their bank card. But your friend tells you that they always repeat the same number for all four digits. How many combinations are there with this condition?
   (b) You have a friend Bob (that is a little better at discrete math) who uses a 4 digit pin number for their bank card. But your friend tells you that they always repeat the same number for three digits. (One example: 5535) How many combinations are there with this condition?

(c) You have a friend Cora (that is not bad at discrete math) who uses a 4 digit pin number for their bank card. But your friend tells you that they always picks two numbers to each repeat twice. (One example: 7007) How many combinations are there with this condition?

(d) You have a friend Dorcas (that is not bad at discrete math) who uses a 4 digit pin number for their bank card. But your friend tells you that they always picks exactly one number to each repeat twice. (One example: 4624) How many combinations are there with this condition?

(e) You have a friend Eugene (that is not bad at discrete math) who uses a 4 digit pin number for their bank card. But your friend tells you that they always four different digits. (One example: 1987) How many combinations are there with this condition?

(f) Which friend has the 'best' method for picking a combination and why?

(4) Let's talk about locks. For a combination lock, the code to unlock the lock is called a combination, not to be confused with *combinations,* $C(n, k)$. **Explain** each of your answers, using complete sentences.

(a) We have a combination lock with a the numbers 0 to 39 on it, spinning wheel that requires a three number code to open, like the ones that were on high school lockers. If we assume the three numbers cannot be the same, how many different combinations to unlock the lock are there?

(b) Using the same combination lock from the previous part, it turns out that is possible for the first and third number to repeat. How many different combinations to unlock the lock are there?

(c) Another lock we have has three spinning 10 digit wheels, like those that are used on briefcases. How many combinations are there to open the lock?

(d) A digital keypad on a door has the digits 0 - 9 on it and requires a 4 digit code to open the door. How many possible combinations are there?

(e) A final digital keypad on a door also has the digits 0 - 9 and requires a 4 digit code to open the door, but you cannot repeat digits more than twice. How many combinations are there? Use your work from the previous pin code problem to make it easier.

(5) In an attempt to clean up your room, you have purchased a new floating shelf to put some of your 17 books you have stacked in a corner. These books are all by different authors. The new book shelf is large enough to hold 10 of the books.

(a) How many ways can you select and arrange 10 of the 17 books on the shelf? Notice that here we will allow the books to end up in any order but each order is considered a different arrangement. Explain.

(b) How many ways can you arrange 10 of the 17 books on the shelf if you insist they must be arranged alphabetically by author? Explain.

(6) Consider functions $f : \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5, 6, 7, 8\}$. **Explain** each of your answers, using complete sentences.

(a) How many functions are there total?

(b) How many functions are injective?

(c) How many functions are surjective?

(d) How many of the injective functions are increasing? To be increasing means that if $a < b$, then $f(a) < f(b)$, or in other words, the outputs get larger as the inputs get larger.